Expert Systems With Applications 301 (2026) 130419

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

Learning-based state transition algorithm

Yingchao Dong [a], Ge Lan [a], Xiaojun Zhou [b,*]

[a] School of Energy Engineering, Xinjiang Institute of Engineering, Urumqi Xinjiang, 830023, China
[b] School of Automation, Central South University, Changsha, 410083, China

ARTICLE INFO

Keywords:
State transition algorithm
Learning mechanism
Metaheuristics
Global optimization

ABSTRACT

The standard state transition algorithm (STA) is an individual-based metaheuristic optimization approach that primarily emphasizes self-learning mechanisms. To extend its adaptability and generalizability, this study proposes a learning-based state transition algorithm (LeSTA) with a population-based framework. In addition to the conventional self-learning strategy, the proposed LeSTA incorporates three novel learning mechanisms: (1) peer-learning, which enables each individual learning from peers within the population; (2) best-learning, which enables individual learning from the best in the population; and (3) experience-learning, which leverages historical search information to guide future optimization directions. Corresponding state transformation operators are meticulously designed to integrate these learning strategies into the algorithm's framework. To validate the effectiveness and superiority of the proposed LeSTA, comprehensive experiments are conducted on a set of benchmark functions and applications. Experimental results have demonstrated the effectiveness and superiority of the proposed LeSTA compared to existing STAs and other state-of-the-art metaheuristic algorithms. The source code for LeSTA will be publicly available at: https://github.com/tiezhongyu2005/LeSTA

1. Introduction

Many practical engineering optimization problems inherently exhibit nonlinear, and non-convex characteristics. Traditional optimization methods, including steepest descent, conjugate gradient, and quasi-Newton techniques, often struggle with such complexities. By contrast, metaheuristic algorithms impose no strict requirements on problem structure. This flexibility has led to broad adoption of two key metaheuristic categories: swarm intelligence-inspired algorithms including, particle swarm optimization (PSO) (Venter & Sobieszczanski-Sobieski, 2003) and ant colony optimization (ACO) (Dorigo et al., 2007), and nature-inspired evolutionary algorithms such as, genetic algorithms (GA) (Holland, 1992) and differential evolution (DE) (Storn & Price, 1997).

The state transition algorithm (STA) (Zhou et al., 2012), as an emerging individual-based metaheuristic, has demonstrated remarkable effectiveness in solving complex optimization problems by conceptualizing solutions as states and treating the solution process as a state transition process. Unlike population-based approaches, standard STA relies on self-learning progressively refining a single solution through state transformations (rotation, expansion, translation, axesion). This mechanism grants STA advantages in computational efficiency and implementation simplicity. It has been widely applied to many engineering fields, such as wind power forecasting (Dong et al., 2021a,b), energy management system optimization, Dong et al. (2024, 2023a,b), feature selection (Du et al., 2023; Zhou et al., 2025), and dynamic optimization (Lin et al., 2022; Zhou et al., 2022), and unmanned aerial vehicle path planning (Zhou et al., 2024), etc. However, its individualistic framework inherently limits information exchange, leading to suboptimal exploration-exploitation balance, susceptibility to local optima, and reduced robustness in high-dimensional spaces.

Recent enhancements to STA primarily focus on three aspects: enhancing parametric adaptability, improving local convergence, and refining the candidate solution update mechanism. For instance, Reference (Zhou et al., 2019) proposed an STA with an optimal parameter selection strategy (POSTA) by utilizing statistical strategies to select the parameter value corresponding to the best fitness function during iteration, while Reference (Dong et al., 2022) further employed statistical strategies to adaptively select state transformation operators of POSTA and utilized quasi-Newton and quadratic interpolation local enhancement techniques to accelerate local convergence in the algorithm's later stages. For the candidate solution update mechanism, Reference (Zhou et al., 2018) proposes a "risk and recovery" strategy. This approach preserves the best solution based on a greedy criterion, while internally accepting inferior solutions with a certain probability, termed risk-taking, and periodically recovering the best solution with another probability. This mechanism significantly enhances the algorithm's likelihood of escaping local optima. In Reference (Han et al., 2019), the idea of

* Corresponding author.
E-mail addresses: dycxju@163.com (Y. Dong), lange19871211@126.com (G. Lan), michael.x.zhou@csu.edu.cn (X. Zhou).

https://doi.org/10.1016/j.eswa.2025.130419
Received 6 September 2025; Received in revised form 23 October 2025; Accepted 11 November 2025
Available online 23 November 2025
0957-4174/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

simulated annealing is introduced into the STA. When the fitness value of the latest state is not better, the state is accepted with a certain probability. The acceptance probability is related to the annealing temperature, that is, the number of iterations of the algorithm, which also increases the possibility of avoiding falling into local optimum. Despite these advances in convergence, parameter tuning, and operator design for solution updates, a critical gap persists: existing variants lack systematic learning mechanisms for leveraging collective intelligence within populations. Population-based frameworks often merely aggregate solutions without enabling effective knowledge transfer between individuals or across iterations.

To bridge this gap, the learning-based state transition algorithm (LeSTA) was proposed in this study. Departing from pure self-learning, LeSTA integrates three novel strategies into a unified population framework.

- Peer-learning: Individuals exchange information with stochastically selected peers, enhancing diversity and escaping local optima.
- Best-learning: Individuals adaptively follow the global best solution, accelerating convergence.
- Experience-learning: Historical search experiences guide future transitions, mitigating premature convergence.

The remainder of this article is as follows. In Section 2, reviews the standard STA. In Section 3, presents the proposed LeSTA in detail. In Section 4, Experiments on 15 benchmark functions and a dynamic economic dispatching problem in a power system demonstrated the effectiveness of LeSTA. Finally, the conclusion is given in Section 5.

2. Standard state transition algorithm

STA is a stochastic optimization method based on the concepts of state space and state transition. In standard STA, a solution of the specific problem is regarded as a state, and the updating process of the incumbent solution is treated as a state transition process. Generally speaking, the uniform form for generating candidate solutions in standard STA can be defined as

$$\begin{cases} \boldsymbol{x}_{t+1} = A_t \boldsymbol{x}_t + B_t \boldsymbol{u}_t \\ y_{t+1} = f(\boldsymbol{x}_{t+1}) \end{cases} \tag{1}$$

where $\boldsymbol{x}_t, \boldsymbol{x}_{t+1} \in R^n$ stand for a current state and the updated state, respectively, which correspond to the solution of a specific problem; $A_t, B_t \in R^{n\times n}$ are state transition matrices; y_t is the fitness value of x_t; $\boldsymbol{u}_t$ is a function of current state $\boldsymbol{x}_t$ and historical states; and f is the fitness function.

2.1. State transformation operators

Within the candidate solution generation framework of the aforementioned STA, the standard STA employs four state transformation operators: rotation transformation, translation transformation, expansion transformation, and axesion transformation (Zhou et al., 2019). These operators are designed to fulfill the needs of global search, local search, and heuristic search. However, the next-generation candidate solutions generated by these transformations depend solely on an individual solution's current state, without leveraging population information sharing or historical solution experience. Consequently, the learning mechanism of the standard STA is limited, characterized as self-learning. This restricts the algorithm's exploration and exploitation of promising regions, thereby constraining its overall optimization performance.

2.2. Regular neighborhood and sampling technique

It is not difficult to understand that for a given solution, an infinite number of candidate solutions can be generated theoretically by using a certain state transformation operator, which can form a regular neighborhood. Obviously, it is not desirable and practical to list all candidate solutions. Considering that the state transition matrix of each operator in STA is randomly generated, the candidate solutions generated by each state transformation are also random, not unique, and any two solutions are independent of each other. Based on the above, the generated candidate solutions can be regarded as samples, and the characteristics of the neighborhood can be reflected by selecting some representative samples. For example, in the axesion transformation, execute the axesion operator SE times independently, and its pseudocode is shown as follows:

```
for i ← 1 to SE do
    State(i, :) ← Best + δR_a Best
end for
```

where SE is the number of samples generated by the axesion transformation, which is denoted as search enforcement. δ is a positive constant, denoted as the axesion factor of the axesion operator. $Best$ is the current best solution.

2.3. Update strategy

As described above, SE candidate solutions can be generated by a particular state transformation operator based on the current best solution. Then, the fitness function value of each candidate solution is calculated, and a new best solution corresponding to the best fitness function value is selected, which is denoted as new_Best. Finally, the current best solution $Best$ is updated using the greedy criterion. The specific update strategy is described as follows:

$$\text{Best} = \begin{cases} \text{new_Best}, & \text{if } f(\text{Best}) > f(\text{new_Best}) \\ \text{Best}, & \text{otherwise} \end{cases} \tag{2}$$

3. Proposed LeSTA

This section introduces the LeSTA, a population-based extension of the standard STA framework. LeSTA retains STA's self-learning capability while integrating three novel learning-based operators: a peer-learning operator for collaborative search, a best-learning operator for accelerated convergence, and an experience-learning operator for historical knowledge utilization. These mechanisms collectively enhance inter-individual knowledge transfer within the population. Furthermore, LeSTA incorporates a crossover mechanism to explicitly regulate exploration-exploitation trade-offs, thereby promoting solution diversity and mitigating premature convergence.

3.1. Learning-based state transformation operators

(1) *Self-learning operators*

Rotation operator

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_t + \alpha \frac{R_r \boldsymbol{x}_t}{n\|\boldsymbol{x}_t\|_2} \tag{3}$$

where $\alpha > 0$, called the rotation factor; $R_r \in \mathbf{R}^{n\times n}$ is a random matrix whose elements obey uniformly distributed in the interval [-1,1]. The rotation transformation ensures that the generated candidate solutions possess local search capability within a hypersphere of maximum radius α, which can be employed to control the precision of the solutions.

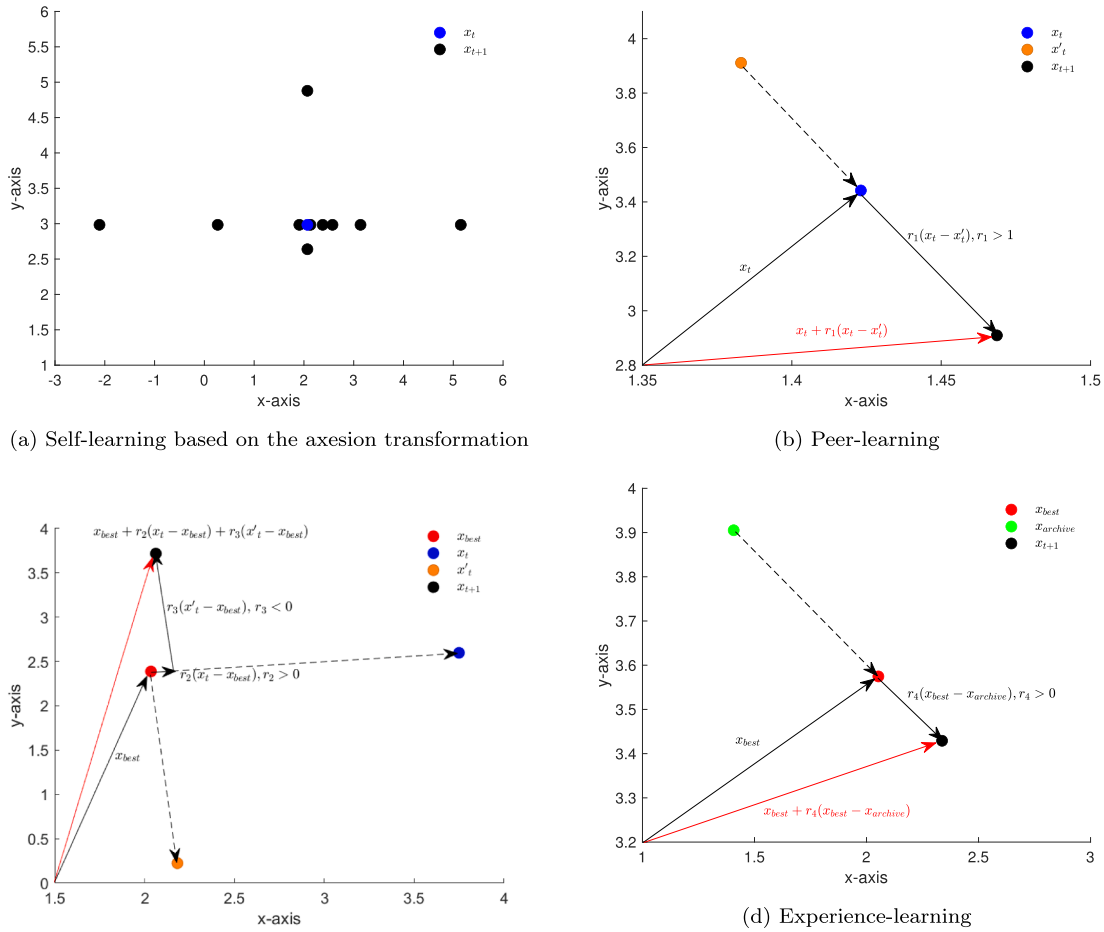
Axesion operator

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_t + \delta R_a \boldsymbol{x}_t, \tag{4}$$

where $\delta > 0$, called the axesion factor; $R_a \in \mathbf{R}^{n\times n}$ is a random matrix whose elements obey Gaussian distribution, and only one random element is nonzero. The axesion transformation is designed to enhance the one-dimensional search capability.

(2) *Peer-learning operator*

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_t + r_1(\boldsymbol{x}_t - \boldsymbol{x}'_t), \tag{5}$$



(a) Self-learning based on the axesion transformation

(b) Peer-learning

(c) Best-learning

(d) Experience-learning

Fig. 1. Four search diagrams based on learning mechanism. (a)Self-learning based on the axesion transformation. (b)Peer-learning. (c)Best-learning. (d)Experience-learning.

where r_1 is a random number following a normal distribution, and $\boldsymbol{x}'_t$ represents an individual in the population distinct from $\boldsymbol{x}_t$. Eq. (5) possesses the functionality of learning from other individuals, thereby enhancing information exchange between different individuals.

(3) *Best-learning operator*

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_{best} + r_2(\boldsymbol{x}_t - \boldsymbol{x}_{best}) + r_3(\boldsymbol{x}'_t - \boldsymbol{x}_{best}), \tag{6}$$

where r_2 and r_3 are random numbers following a normal distribution, and $\boldsymbol{x}'_t$ and $\boldsymbol{x}_{best}$ represent other individuals within the population and the best individual in the population, respectively. Eq. (6) indicates that individuals in the population learn from the best individual to accelerate the convergence speed of the population.

(4) *Experience-learning operator*

$$\boldsymbol{x}_{t+1} = \boldsymbol{x}_{best} + r_4(\boldsymbol{x}_{best} - \boldsymbol{x}_{archive}), \tag{7}$$

where r_4 is a random number following a uniform distribution over [-1, 1], and $\boldsymbol{x}_{archive}$ represents one from the historical best solution set. Eq. (7) indicates that the current best solution learns from historical experience, thus enhancing the diversity of the next generation's population.

Using a two-dimensional search space as an example, Fig. 1 illustrates the schematic diagrams of self-learning based on the axesion transformation, peer-learning, best-learning, and experience-learning. The schematic diagram of the rotation transformation can be found in Dong et al. (2022).

3.2. Stochastic crossover operator

To balance the exploration and exploitation capabilities of the algorithm, a stochastic crossover operator is introduced in LeSTA. This promotes solution diversity and mitigates premature convergence. It is worth emphasizing that the stochastic crossover is applied solely after the peer-learning, best-learning, and experience-learning operators. The definition of the stochastic crossover operator is as follows:

$$\widetilde{x}_{d,t+1} = \begin{cases} x_{d,t+1}, & \text{if } \mathrm{rand}_j(0,1] \le C_r \\ x_{d,t}, & \text{otherwise} \end{cases} \tag{8}$$

where d denotes the dth dimension; C_r is the crossover probability, which is a random number within [0,1] during the tth iteration; and $\widetilde{x}_{d,t+1}$ represents the position of the dth dimension in the offspring individual after performing the crossover operation.

3.3. Framework of LeSTA

Integrating the aforementioned components, Algorithm 1 presents the complete pseudocode of the proposed LeSTA. where Archive stores historical best solutions, State represents the population positions, and fState denotes the fitness values of the population.

The flowchart of the LeSTA is outlined in Fig. 2.

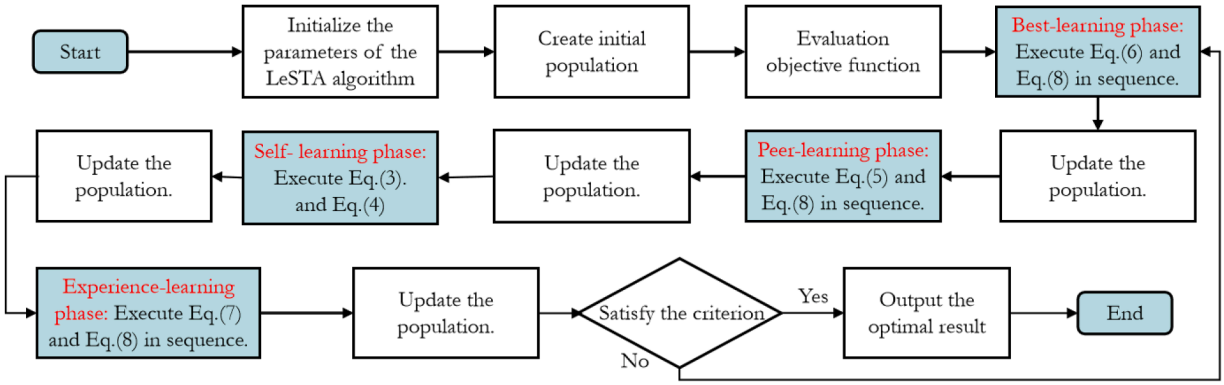


Fig. 2. The flowchart of LeSTA.

Algorithm 1 Pseudocode of LeSTA.

Require: func (objective function); N (population size); SE; δ
Ensure: $Best$: the optimal solution
State ← RandomInitialize(·)
repeat
  if $\alpha < \alpha_{\min}$ **then**
    $\alpha \leftarrow \alpha_{\max}$
  end if
  [State,fState,Archive] ← Best_learning(func,State,Archive,N, SE)
  [State,fState,Archive] ← Peer_learning(func,State,Archive,N, SE)
  [State,fState,Archive] ← Self_learning(func,State,Archive,N, SE)
  [State,fState,Archive] ← Experience_learning(func,State,Archive, N, SE)
  [fBest,index] = min(fState);
  Best = State(index,:);
  $\alpha \leftarrow \frac{\alpha}{fc}$
until the specified termination criterion is met

4. Results and discussion

4.1. Experimental setup

To evaluate the performance of the proposed LeSTA, 15 widely used benchmark functions from Dong et al. (2022) were adopted. Following the practice in Kudela (2022), the global optima of these functions were shifted to non-zero values to prevent potential algorithm bias toward the origin. The detailed specifications of these modified functions are provided in Table 1. The results of LeSTA were compared with 12 other metaheuristic algorithms, categorized into two primary groups: the STA family and several other popular/state-of-the-art metaheuristics. The STA family includes STA (Zhou et al., 2012), DaSTA (Zhou et al., 2018), STASA (Han et al., 2019), POSTA (Zhou et al., 2019), and ASTA (Dong et al., 2022). The other metaheuristics comprise GWO (Mirjalili et al., 2014), HHO (Heidari et al., 2019), RUN (Ahmadianfar et al., 2021), ESC (Ouyang et al., 2024), CEO (Dong et al., 2025), AOO (Wang et al., 2025), and PGA (Bohat et al., 2025). Among these, GWO, HHO, and RUN serve as well-established baselines, having been cited over 20,000, 5,800, and 900 times, respectively.

In the experiments, the dimensionality (n) of the test functions was set to 30 and 50. All algorithms utilized a population size of 30 and a maximum number of function evaluations (MaxFES) of $3e3 \times n \times \log_2(n)$. To ensure fair comparison, parameter settings for the compared algorithms followed their original publications. For LeSTA, the SE parameter was set to 10, $\alpha_{\min} = 1e-4, \alpha_{\max} = 1, fc = 2$, and $\delta = 1$. All algorithms were independently executed 30 times on a Windows 10 desktop with an Intel® Core™ i5-9500F CPU @ 3.00 GHz and 16 GB RAM, using MATLAB R2020b.

4.2. Comparisons with STA variants and other metaheuristic algorithms

To rigorously evaluate the proposed LeSTA, comprehensive comparative experiments were conducted against 12 optimization techniques across 15 benchmark functions at dimensionalities of 30 and 50. Experimental results unequivocally demonstrate that LeSTA achieves statistically superior performance in solution accuracy, convergence stability, and scalability across virtually all test problems. The experimental results are shown in Tables 2 and 3.

As can be seen from Tables 2 and 3, LeSTA demonstrates highly competitive performance on unimodal functions ($F_1 - F_8$), surpassing most competitors while showing nuanced differences with the CEO algorithm. For instance, on the 30D Sphere_shifed function designated as F_1, CEO achieves the best recorded mean fitness of 4.05×10^{-30}, slightly outperforming LeSTA's 1.10×10^{-28} by approximately two orders of magnitude. Nevertheless, this competitive pattern shifts significantly at multimodal functions ($F_9 - F_{15}$), as evidenced on function F_9 at 50D where LeSTA's solution quality of 4.60×10^{-27} exceeds CEO's 4.78×10^{-2} by approximately 25 orders of magnitude while also substantially outperforming ASTA at 3.82×10^{-22} and POSTA at 2.97×10^{-21}. LeSTA's advantage becomes more pronounced in complex multimodal landscapes where competing methods frequently stagnate. On the 50D function F_{15}, LeSTA's solution of 1.17×10^{-12} exceeds CEO's 1.46×10^{-3} by nine orders of magnitude while dramatically outperforming POSTA at 2.20×10^{-6}, demonstrating enhanced capability in escaping deceptive local basins where other methods become trapped. Furthermore, LeSTA consistently achieves significantly smaller standard deviations across both unimodal and multimodal functions compared to most competing algorithms, indicating superior stability and reliability in its optimization performance.

The foundational superiority of LeSTA stems from its novel population-based architecture that synergistically integrates four complementary strategies. Best-learning accelerates convergence by adaptively steering individuals toward the global optimum, enabling high-precision solutions for unimodal functions. Peer-learning enhances exploration diversity through stochastic information exchange among population members, effectively preventing premature convergence in multimodal landscapes. Experience-learning leverages historical search trajectories to guide optimization directions, conferring stability in high-dimensional spaces. The retention of STA's self-learning operator facilitates effective exploitation of promising locations within individual neighborhoods. This framework directly addresses critical limitations inherent in traditional STA variants: the absence of collective intelligence mechanisms, excessive dependence on self-learning operators

Table 1
Fifteen well-known benchmark functions.

Name	Functions	Range	x^*	$f(x^*)$
Spherical_shifted	$F_1(x) = f_1(x-s), s = 2:n+1$ $f_1(x) = \sum_{i=1}^{n} x_i^2$	$[0, 2n]$	$[2, \ldots, n+1]$	0
Schwefel2.22_shifted	$F_2(x) = f_2(x-s), s = 2:n+1$ $f_2(x) = \sum_{i=1}^{n} \lvert x_i \rvert + \prod_{i=1}^{n} \lvert x_i \rvert$	$[0, 2n]$	$[2, \ldots, n+1]$	0
Schwefel1.2_shifted	$F_3(x) = f_3(x-s), s = 2:n+1$ $f_3(x) = \sum_{i=1}^{n} \left(\sum_{j=1}^{i} x_j \right)^2$	$[0, 2n]$	$[2, \ldots, n+1]$	0
Rosenbrock_shifted	$F_4(x) = f_4(x-s), s = 2:n+1$ $f_4(x) = \sum_{i=1}^{n} \left(100\left(x_{i+1} - x_i^2\right)^2 + \left(x_i - 1\right)^2 \right)$	$[0, 2n]$	$[3, \ldots, n+2]$	0
Schwefel2.4_shifted	$F_5(x) = f_5(x-s), s = 2:n+1$ $f_5(x) = \sum_{i=1}^{n} \left[(x_i - 1)^2 + (x_1 - x_i^2)^2 \right]$	$[0, 2n]$	$[3, \ldots, n+2]$	0
Elliptic_shifted	$F_6(x) = f_6(x-s), s = 2:n+1$ $f_6(x) = \sum_{i=1}^{n} \left(10^6\right)^{\frac{i-1}{n-1}} x_i^2$	$[0, 2n]$	$[2, \ldots, n+1]$	0
Tablet_shifted	$F_7(x) = f_7(x-s), s = 2:n+1$ $f_7(x) = 10^6 \cdot x_1^2 + \sum_{i=2}^{n} x_i^6$	$[0, 2n]$	$[2, \ldots, n+1]$	0
Zakharov_shifted	$F_8(x) = f_8(x-s), s = 2:n+1$ $f_8(x) = \sum_{i=1}^{n} x_i^2 + \left(\sum_{i=1}^{n} 0.5ix_i \right)^2 + \left(\sum_{i=1}^{n} 0.5ix_i \right)^4$	$[0, 2n]$	$[2, \ldots, n+1]$	0
Penalized1_shifted	$F_9(x) = f_9(x-s), s = 2:n+1$ $f_9(x) = \frac{\pi}{n} \left\{ \sum_{i=1}^{n-1} \left(y_i - 1\right)^2 [1 + 10\sin^2\left(\pi y_{i+1}\right)] \right.$ $\left. + \left(y_n - 1\right)^2 + 10\sin\left(\pi y_1\right) \right\} + \sum_{i=1}^{n} u\left(x_i, 10, 100, 4\right)$ $u\left(x_i, a, k, m\right) = \begin{cases} k\left(x_i - a\right)^m x_i > a \\ 0 - a < x_i < a \\ k\left(-x_i - a\right)^m x_i < a \end{cases}$	$[0, 2n]$	$[1, \ldots, n]$	0
Penalized2_shifted	$F_{10}(x) = f_{10}(x-s), s = 2:n+1$ $f_{10}(x) = 0.1 \left\{ \sum_{i=1}^{n-1} \left(x_i - 1\right)^2 \left[1 + \sin^2\left(3\pi x_i + 1\right)\right] + \right.$ $\left(x_n - 1\right)^2 \left[1 + \sin^2\left(2\pi x_n\right)\right] +$ $\left. \sin^2\left(3\pi x_1\right) \right\} + \sum_{i=1}^{n} u\left(x_i, 5, 100, 4\right)$	$[0, 2n]$	$[3, \ldots, n+2]$	0
Ackley_shifted	$F_{11}(x) = f_{11}(x-s), s = 2:n+1$ $f_{11}(x) = 20 + e - 20\exp\left(-0.2\sqrt{\frac{1}{n} \sum_{i=1}^{n} x_i^2} \right)$ $- \exp\left(\frac{1}{n} \sum_{i=1}^{n} \cos\left(2\pi x_i\right) \right)$	$[0, 2n]$	$[2, \ldots, n+1]$	0
Griewank_shifted	$F_{12}(x) = f_{12}(x-s), s = 2:n+1$ $f_{12}(x) = \frac{1}{4000} \sum_{i=1}^{n} x_i^2 - \prod_{i=1}^{n} \cos(\frac{x_i}{\sqrt{i}}) + 1$	$[0, 2n]$	$[2, \ldots, n+1]$	0
Rastrigin_shifted	$F_{13}(x) = f_{13}(x-s), s = 2:n+1$ $f_{13}(x) = \sum_{i=1}^{n} \left(x_i^2 - 10\cos\left(2\pi x_i\right) + 10\right)$	$[0, 2n]$	$[2, \ldots, n+1]$	0
Levy&Montalvo1_shifted	$F_{14}(x) = f_{14}(x-s), s = 2:n+1$ $f_{14}(x) = \frac{\pi}{n}(10\sin^2\left(\pi y_1\right) + \sum_{n-1}^{i=1} \left(y_i - 1\right)^2 [1 + 10\sin^2\left(\pi y_{i+1}\right)]$ $+\left(y_n - 1\right)^2), y_i = 1 + \frac{1}{4}\left(x_i + 1\right)$	$[0, 2n]$	$[1, \ldots, n]$	0
Levy&Montalvo2_shifted	$F_{15}(x) = f_{15}(x-s), s = 2:n+1$ $f_{15}(x) = 0.1 \left(\sum_{i=1}^{n-1} \left(x_i - 1\right)^2 \left(1 + \left(\sin^2\left(3\pi x_{i+1}\right)\right)\right) \right.$ $\left. +\left(x_n - 1\right)^2 \left(1 + \left(\sin^2\left(2\pi x_n\right)\right)\right) + \left(\sin^2\left(3\pi x_1\right)\right) \right)$	$[0, 2n]$	$[3, \ldots, n+2]$	0

leading to exploration-exploitation imbalances, and dimensional sensitivity caused by insufficient utilization of historical search information.

4.3. Statistical analysis

To further demonstrate the significant superiority of LeSTA over the other 12 compared algorithms, this subsection conducts two nonparametric statistical analyses based on the results of 30 independent runs, namely the Friedman test (Pereira et al., 2015) and the Wilcoxon rank-sum test (Derrac et al., 2011). Fig. 3 presents the average rankings of the 13 algorithms on 15 benchmark functions obtained from the Friedman test, while Fig. 4a and b illustrate the pairwise Wilcoxon rank-sum test results for the 30D and 50D test functions, respectively. For a more detailed function-by-function comparison, the complete Wilcoxon

Table 2
Comparison of optimization algorithms on fifteen well-known benchmark functions (D = 30).

Func	Index	GWO	HHO	RUN	ESC	CEO	AOO	PGA	STA	DaSTA	STASA	POSTA	ASTA	LeSTA
F_1	mean	2.54×10^{2}	4.19×10^{0}	6.48×10^{-5}	2.75×10^{-5}	4.05×10^{-30}	8.56×10^{-11}	2.20×10^{0}	8.69×10^{-11}	2.26×10^{-18}	1.04×10^{-15}	4.54×10^{-19}	1.79×10^{-20}	1.10×10^{-28}
	std	2.03×10^{2}	8.38×10^{-1}	3.72×10^{-5}	1.05×10^{-5}	6.83×10^{-30}	4.97×10^{-11}	6.73×10^{0}	1.50×10^{-11}	3.44×10^{-19}	4.65×10^{-16}	7.80×10^{-20}	7.32×10^{-21}	2.08×10^{-28}
F_2	mean	9.02×10^{1}	2.43×10^{2}	5.43×10^{1}	1.68×10^{-2}	1.13×10^{-15}	3.96×10^{1}	3.07×10^{0}	1.28×10^{-2}	1.34×10^{-8}	8.14×10^{-3}	3.39×10^{-9}	2.52×10^{-9}	1.03×10^{-13}
	std	3.52×10^{1}	3.00×10^{1}	8.94×10^{1}	3.36×10^{-3}	3.26×10^{-15}	8.23×10^{1}	5.63×10^{0}	8.26×10^{-3}	6.18×10^{-9}	6.00×10^{-3}	5.50×10^{-10}	1.27×10^{-9}	2.39×10^{-13}
F_3	mean	6.64×10^{2}	4.43×10^{1}	7.95×10^{-2}	1.00×10^{-3}	7.03×10^{-14}	9.72×10^{-5}	2.83×10^{2}	1.53×10^{-7}	4.87×10^{-5}	2.16×10^{-5}	7.18×10^{-12}	6.51×10^{-12}	2.59×10^{-15}
	std	3.14×10^{2}	8.91×10^{0}	2.66×10^{-1}	3.28×10^{-4}	8.58×10^{-14}	5.08×10^{-5}	2.99×10^{2}	9.19×10^{-8}	3.78×10^{-5}	9.98×10^{-6}	9.16×10^{-12}	7.79×10^{-12}	4.66×10^{-15}
F_4	mean	2.68×10^{6}	8.30×10^{2}	3.63×10^{1}	3.28×10^{1}	1.83×10^{-9}	6.76×10^{1}	1.55×10^{4}	3.27×10^{1}	1.84×10^{1}	4.67×10^{1}	1.16×10^{1}	2.11×10^{1}	1.24×10^{-6}
	std	4.30×10^{6}	5.32×10^{2}	1.94×10^{1}	3.07×10^{1}	7.56×10^{-9}	7.49×10^{1}	4.83×10^{4}	3.40×10^{1}	1.14×10^{1}	3.69×10^{1}	1.31×10^{1}	2.33×10^{1}	6.68×10^{-6}
F_5	mean	6.21×10^{4}	2.04×10^{1}	8.10×10^{-5}	2.35×10^{-4}	1.65×10^{-27}	5.17×10^{-9}	1.67×10^{3}	5.35×10^{-10}	2.63×10^{-15}	1.02×10^{-14}	3.21×10^{-18}	4.01×10^{-19}	2.88×10^{-27}
	std	1.06×10^{5}	4.00×10^{0}	2.71×10^{-5}	9.44×10^{-5}	9.70×10^{-28}	6.09×10^{-9}	5.19×10^{3}	6.63×10^{-11}	2.92×10^{-15}	2.55×10^{-14}	7.43×10^{-19}	1.78×10^{-19}	8.91×10^{-27}
F_6	mean	3.68×10^{6}	1.73×10^{5}	2.47×10^{3}	4.63×10^{1}	7.92×10^{-27}	3.10×10^{4}	2.61×10^{2}	1.32×10^{-2}	1.81×10^{-13}	3.34×10^{-2}	1.01×10^{-14}	3.97×10^{-15}	2.90×10^{-23}
	std	4.66×10^{6}	5.19×10^{4}	8.59×10^{2}	7.89×10^{1}	3.28×10^{-26}	9.38×10^{3}	8.68×10^{2}	1.11×10^{-2}	9.39×10^{-14}	2.71×10^{-2}	3.39×10^{-15}	2.68×10^{-15}	9.31×10^{-23}
F_7	mean	5.78×10^{6}	1.04×10^{2}	1.89×10^{-3}	2.29×10^{-13}	4.99×10^{-87}	2.21×10^{-2}	1.01×10^{5}	1.56×10^{-10}	9.18×10^{-21}	5.03×10^{-9}	1.12×10^{-21}	3.75×10^{-22}	3.27×10^{-84}
	std	1.03×10^{7}	7.25×10^{1}	9.20×10^{-4}	1.77×10^{-13}	2.30×10^{-86}	1.74×10^{-2}	5.36×10^{5}	3.12×10^{-10}	2.82×10^{-20}	7.95×10^{-9}	1.28×10^{-21}	5.75×10^{-22}	1.68×10^{-83}
F_8	mean	1.86×10^{3}	3.35×10^{1}	5.64×10^{-3}	9.54×10^{-5}	4.31×10^{-19}	3.82×10^{-6}	1.45×10^{3}	6.25×10^{-10}	3.32×10^{-10}	4.42×10^{-8}	1.81×10^{-18}	3.10×10^{-18}	4.13×10^{-19}
	std	1.03×10^{3}	9.54×10^{0}	3.75×10^{-3}	3.30×10^{-5}	9.08×10^{-19}	2.26×10^{-6}	8.70×10^{2}	2.16×10^{-10}	6.19×10^{-10}	5.47×10^{-8}	3.56×10^{-19}	1.33×10^{-18}	2.10×10^{-18}
F_9	mean	9.74×10^{4}	7.75×10^{0}	2.17×10^{-1}	5.46×10^{-1}	7.62×10^{-2}	6.91×10^{-3}	9.95×10^{4}	8.71×10^{-13}	1.92×10^{-19}	1.54×10^{-14}	4.52×10^{-21}	7.31×10^{-22}	1.10×10^{-29}
	std	3.87×10^{5}	2.71×10^{0}	1.02×10^{0}	8.44×10^{-1}	1.90×10^{-1}	2.59×10^{-2}	5.12×10^{5}	1.53×10^{-13}	6.55×10^{-19}	6.75×10^{-15}	8.34×10^{-22}	2.74×10^{-22}	2.97×10^{-29}
F_{10}	mean	4.81×10^{5}	8.72×10^{-1}	5.10×10^{-3}	5.60×10^{-1}	7.32×10^{-4}	3.63×10^{-3}	7.08×10^{4}	1.35×10^{-11}	5.52×10^{-19}	6.79×10^{-15}	7.06×10^{-20}	1.00×10^{-20}	1.02×10^{-28}
	std	1.40×10^{6}	1.85×10^{-1}	6.09×10^{-3}	2.05×10^{0}	2.74×10^{-3}	5.81×10^{-3}	3.72×10^{5}	1.95×10^{-12}	1.29×10^{-19}	3.58×10^{-15}	1.10×10^{-20}	4.16×10^{-21}	3.58×10^{-28}
F_{11}	mean	1.13×10^{1}	1.33×10^{1}	1.58×10^{1}	1.40×10^{0}	6.72×10^{-15}	3.11×10^{-2}	9.21×10^{0}	6.92×10^{-6}	1.17×10^{-9}	3.06×10^{-9}	5.21×10^{-10}	9.21×10^{-11}	2.82×10^{-14}
	std	2.96×10^{0}	5.47×10^{0}	6.71×10^{0}	1.00×10^{0}	1.50×10^{-15}	1.67×10^{-1}	6.21×10^{0}	4.70×10^{-7}	1.27×10^{-10}	2.12×10^{-9}	4.07×10^{-11}	1.45×10^{-11}	4.19×10^{-14}
F_{12}	mean	8.91×10^{-1}	1.85×10^{-1}	9.72×10^{-2}	2.03×10^{-2}	1.29×10^{-2}	6.83×10^{-2}	2.91×10^{-1}	1.72×10^{-1}	2.53×10^{-2}	1.52×10^{-1}	2.13×10^{-1}	2.71×10^{-2}	1.00×10^{-2}
	std	1.93×10^{-1}	3.46×10^{-2}	8.62×10^{-2}	2.68×10^{-2}	9.99×10^{-3}	1.06×10^{-1}	3.08×10^{-1}	2.47×10^{-1}	6.59×10^{-2}	2.42×10^{-1}	2.71×10^{-1}	2.41×10^{-2}	1.25×10^{-2}
F_{13}	mean	3.14×10^{2}	4.16×10^{2}	1.35×10^{2}	3.75×10^{1}	9.92×10^{0}	8.27×10^{1}	1.34×10^{2}	2.12×10^{-8}	1.00×10^{-13}	3.98×10^{-14}	1.33×10^{-1}	9.66×10^{-14}	7.39×10^{-14}
	std	1.42×10^{2}	1.04×10^{2}	6.09×10^{1}	2.87×10^{1}	3.55×10^{0}	2.15×10^{1}	9.40×10^{1}	3.57×10^{-9}	4.80×10^{-14}	1.83×10^{-13}	7.14×10^{-1}	3.92×10^{-14}	1.11×10^{-13}
F_{14}	mean	7.19×10^{0}	1.99×10^{1}	5.51×10^{-1}	2.76×10^{-2}	6.22×10^{-2}	1.73×10^{-2}	3.12×10^{-1}	9.26×10^{-13}	3.17×10^{-19}	2.13×10^{-14}	4.24×10^{-21}	6.41×10^{-22}	3.09×10^{-29}
	std	4.14×10^{0}	5.88×10^{0}	2.87×10^{0}	5.94×10^{-2}	9.50×10^{-2}	5.41×10^{-2}	4.36×10^{-1}	1.37×10^{-13}	9.69×10^{-19}	8.41×10^{-15}	8.49×10^{-22}	2.28×10^{-22}	9.31×10^{-29}
F_{15}	mean	8.21×10^{1}	1.35×10^{2}	7.49×10^{-3}	8.79×10^{-3}	3.66×10^{-4}	1.81×10^{0}	1.17×10^{0}	9.42×10^{2}	2.75×10^{0}	3.37×10^{2}	1.63×10^{-9}	1.27×10^{-1}	3.44×10^{-4}
	std	4.72×10^{1}	1.05×10^{2}	9.00×10^{-3}	3.14×10^{-2}	1.97×10^{-3}	4.26×10^{0}	2.87×10^{0}	3.67×10^{1}	1.48×10^{1}	1.78×10^{2}	8.24×10^{-9}	3.65×10^{-1}	1.85×10^{-3}

Table 3
Comparison of optimization algorithms on fifteen well-known benchmark functions (D = 50).

Func	Index	GWO	HHO	RUN	ESC	CEO	AOO	PGA	STA	DaSTA	STASA	POSTA	ASTA	LeSTA
F_1	mean	2.51×10^{3}	1.85×10^{1}	1.69×10^{-3}	1.88×10^{-4}	1.28×10^{-28}	2.92×10^{-10}	1.74×10^{2}	8.77×10^{-11}	2.24×10^{-18}	1.86×10^{-18}	5.57×10^{-19}	2.14×10^{-20}	2.83×10^{-25}
	std	1.30×10^{3}	2.35×10^{0}	7.77×10^{-4}	5.02×10^{-5}	8.15×10^{-29}	1.24×10^{-10}	2.55×10^{2}	7.92×10^{-12}	3.44×10^{-19}	2.07×10^{-19}	8.86×10^{-20}	6.48×10^{-21}	1.17×10^{-24}
F_2	mean	4.24×10^{2}	7.13×10^{2}	5.05×10^{11}	6.07×10^{-2}	1.38×10^{-14}	9.30×10^{4}	3.59×10^{1}	4.11×10^{-2}	2.94×10^{-8}	2.80×10^{-2}	5.91×10^{-9}	5.80×10^{-9}	1.24×10^{-8}
	std	9.60×10^{1}	8.41×10^{1}	2.37×10^{12}	7.08×10^{-3}	1.03×10^{-14}	4.88×10^{5}	4.51×10^{1}	1.58×10^{-2}	8.62×10^{-9}	1.16×10^{-2}	1.48×10^{-9}	2.32×10^{-9}	6.52×10^{-8}
F_3	mean	6.65×10^{3}	5.08×10^{2}	9.40×10^{0}	2.57×10^{-2}	1.10×10^{-8}	7.32×10^{-3}	1.86×10^{3}	1.43×10^{-4}	2.34×10^{-3}	1.77×10^{-3}	3.09×10^{-7}	3.54×10^{-7}	4.85×10^{-7}
	std	2.86×10^{3}	1.10×10^{2}	1.91×10^{1}	6.27×10^{-3}	9.76×10^{-9}	2.45×10^{-3}	1.16×10^{3}	4.87×10^{-5}	9.15×10^{-4}	6.81×10^{-4}	2.33×10^{-7}	3.21×10^{-7}	1.09×10^{-6}
F_4	mean	1.06×10^{8}	4.53×10^{3}	5.89×10^{1}	8.40×10^{1}	3.05×10^{-4}	6.68×10^{1}	2.54×10^{7}	6.46×10^{1}	3.97×10^{1}	9.04×10^{1}	3.72×10^{1}	4.01×10^{1}	2.27×10^{-5}
	std	1.10×10^{8}	1.08×10^{3}	2.54×10^{1}	7.76×10^{1}	1.64×10^{-3}	3.92×10^{1}	4.58×10^{7}	4.75×10^{1}	1.63×10^{1}	4.94×10^{1}	4.34×10^{1}	2.97×10^{1}	1.18×10^{-4}
F_5	mean	1.36×10^{6}	9.18×10^{1}	3.10×10^{-3}	1.53×10^{-3}	2.99×10^{-26}	9.03×10^{-9}	1.80×10^{5}	5.64×10^{-10}	3.19×10^{-15}	1.51×10^{-16}	3.34×10^{-18}	5.66×10^{-19}	5.60×10^{-25}
	std	1.49×10^{6}	2.01×10^{1}	1.06×10^{-3}	4.62×10^{-4}	1.44×10^{-26}	1.04×10^{-8}	3.45×10^{5}	5.91×10^{-11}	1.43×10^{-15}	2.15×10^{-16}	5.34×10^{-19}	1.96×10^{-19}	2.75×10^{-24}
F_6	mean	6.15×10^{7}	9.95×10^{5}	2.05×10^{4}	1.42×10^{4}	8.10×10^{-24}	1.27×10^{5}	2.44×10^{4}	4.32×10^{-2}	4.03×10^{-13}	9.52×10^{-2}	2.03×10^{-14}	1.16×10^{-14}	4.09×10^{-17}
	std	4.23×10^{7}	1.99×10^{5}	5.45×10^{3}	6.09×10^{3}	2.83×10^{-23}	4.17×10^{4}	6.61×10^{4}	3.22×10^{-2}	1.77×10^{-13}	5.62×10^{-2}	5.64×10^{-15}	5.03×10^{-15}	2.18×10^{-16}
F_7	mean	9.06×10^{8}	8.08×10^{2}	1.67×10^{-3}	3.55×10^{-12}	3.74×10^{-83}	1.98×10^{-2}	1.64×10^{8}	1.31×10^{-9}	7.74×10^{-22}	1.03×10^{-8}	2.60×10^{-22}	7.93×10^{-23}	8.41×10^{-83}
	std	1.16×10^{9}	4.03×10^{2}	6.52×10^{-4}	2.33×10^{-12}	1.31×10^{-82}	8.37×10^{-3}	4.00×10^{8}	1.60×10^{-9}	1.35×10^{-21}	2.27×10^{-8}	4.96×10^{-22}	1.54×10^{-22}	3.60×10^{-82}
F_8	mean	1.21×10^{4}	1.05×10^{3}	4.51×10^{-1}	1.53×10^{-3}	1.59×10^{-8}	1.34×10^{-4}	1.36×10^{4}	5.99×10^{-1}	1.78×10^{-5}	1.07×10^{-1}	3.72×10^{-16}	4.54×10^{-17}	5.40×10^{-8}
	std	4.81×10^{3}	3.97×10^{2}	4.03×10^{-1}	7.27×10^{-4}	2.60×10^{-8}	5.07×10^{-5}	6.89×10^{3}	1.93×10^{0}	1.32×10^{-5}	2.01×10^{-1}	7.12×10^{-16}	2.16×10^{-17}	1.21×10^{-7}
F_9	mean	2.90×10^{7}	1.24×10^{1}	2.67×10^{-1}	2.02×10^{1}	4.78×10^{-2}	2.28×10^{-2}	1.19×10^{6}	4.82×10^{-13}	2.32×10^{-20}	1.54×10^{-20}	2.97×10^{-21}	3.82×10^{-22}	4.60×10^{-27}
	std	3.62×10^{7}	3.08×10^{0}	8.21×10^{-1}	1.32×10^{1}	1.23×10^{-1}	6.11×10^{-2}	2.78×10^{6}	5.67×10^{-14}	9.85×10^{-21}	1.94×10^{-21}	3.14×10^{-22}	1.28×10^{-22}	2.33×10^{-26}
F_{10}	mean	9.39×10^{7}	6.27×10^{0}	2.29×10^{1}	7.40×10^{1}	1.83×10^{-3}	1.83×10^{-3}	1.40×10^{7}	1.24×10^{-11}	4.54×10^{-19}	3.32×10^{-19}	7.46×10^{-20}	8.04×10^{-21}	1.22×10^{-27}
	std	9.21×10^{7}	1.17×10^{1}	1.47×10^{1}	1.96×10^{1}	4.09×10^{-3}	4.09×10^{-3}	2.51×10^{7}	1.65×10^{-12}	8.92×10^{-20}	4.55×10^{-20}	7.20×10^{-21}	1.94×10^{-21}	2.39×10^{-27}
F_{11}	mean	2.02×10^{1}	1.89×10^{1}	1.99×10^{1}	2.00×10^{1}	8.23×10^{-1}	2.00×10^{1}	1.99×10^{1}	5.60×10^{-6}	1.01×10^{-9}	1.92×10^{-7}	4.66×10^{0}	8.38×10^{-11}	9.30×10^{-12}
	std	3.02×10^{-1}	9.77×10^{-3}	5.72×10^{-3}	2.63×10^{-2}	3.64×10^{0}	1.45×10^{-10}	1.21×10^{-2}	2.99×10^{-7}	8.79×10^{-11}	4.03×10^{-7}	8.44×10^{0}	1.00×10^{-11}	4.82×10^{-11}
F_{12}	mean	1.61×10^{0}	4.50×10^{-1}	1.10×10^{-2}	1.06×10^{-1}	1.29×10^{-2}	1.01×10^{-1}	4.93×10^{-1}	8.13×10^{-2}	4.09×10^{-2}	7.06×10^{-3}	2.94×10^{-1}	2.27×10^{-2}	8.53×10^{-3}
	std	2.39×10^{-1}	5.49×10^{-2}	3.99×10^{-2}	1.09×10^{-1}	1.25×10^{-2}	2.13×10^{-1}	2.86×10^{-1}	2.13×10^{-1}	1.10×10^{-1}	3.80×10^{-2}	2.89×10^{-1}	2.31×10^{-2}	2.26×10^{-2}
F_{13}	mean	2.76×10^{3}	9.81×10^{2}	3.81×10^{2}	1.15×10^{3}	1.64×10^{1}	1.66×10^{2}	5.83×10^{2}	3.32×10^{-2}	2.08×10^{-13}	6.63×10^{-2}	3.26×10^{-13}	6.84×10^{-13}	4.13×10^{-13}
	std	9.12×10^{2}	2.60×10^{2}	1.43×10^{2}	3.85×10^{2}	5.33×10^{0}	4.21×10^{1}	4.35×10^{2}	1.79×10^{-1}	1.02×10^{-13}	2.48×10^{-1}	2.10×10^{-13}	1.95×10^{-13}	5.87×10^{-13}
F_{14}	mean	5.76×10^{1}	6.21×10^{1}	2.78×10^{1}	1.99×10^{0}	3.94×10^{-2}	7.78×10^{0}	5.71×10^{0}	4.78×10^{-13}	5.00×10^{-19}	1.48×10^{-20}	2.90×10^{-21}	3.96×10^{-22}	2.38×10^{-28}
	std	2.34×10^{1}	8.07×10^{0}	2.43×10^{1}	2.10×10^{0}	5.90×10^{-2}	1.09×10^{1}	6.99×10^{0}	6.10×10^{-14}	2.50×10^{-18}	1.82×10^{-21}	3.92×10^{-22}	1.41×10^{-22}	6.00×10^{-28}
F_{15}	mean	6.70×10^{2}	1.28×10^{3}	1.44×10^{2}	7.19×10^{0}	1.46×10^{-3}	3.03×10^{1}	1.75×10^{1}	4.36×10^{3}	5.04×10^{1}	1.60×10^{3}	2.20×10^{-6}	7.97×10^{1}	1.17×10^{-12}
	std	2.39×10^{2}	2.20×10^{2}	1.37×10^{2}	2.06×10^{1}	7.89×10^{-3}	3.27×10^{1}	2.72×10^{1}	9.49×10^{1}	2.17×10^{2}	1.18×10^{3}	1.19×10^{-5}	3.13×10^{2}	5.58×10^{-12}

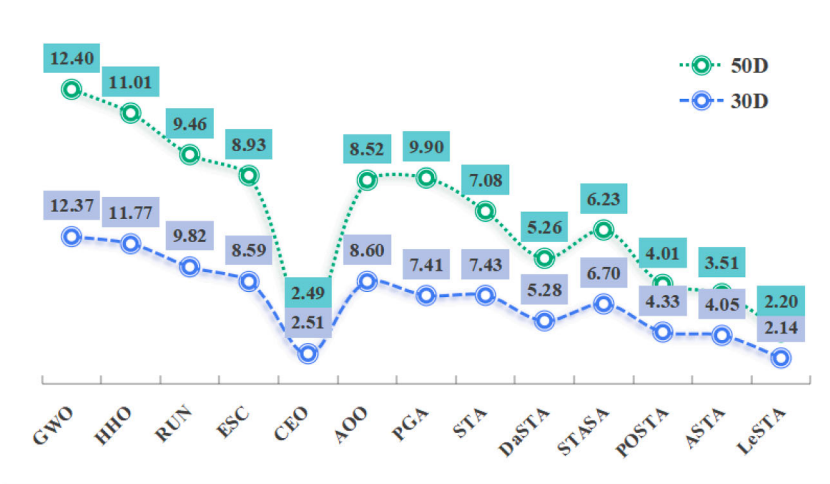


Fig. 3. Friedman test ranking results of thirteen algorithms on fifteen test functions.

Table 4
Wilcoxon rank-sum test results of LeSTA versus other algorithms on 30-dimensional functions.

Func	GWO	HHO	RUN	ESC	CEO	AOO	PGA	STA	DaSTA	STASA	POSTA	ASTA
F_1	−	−	−	−	+	−	−	−	−	−	−	−
F_2	−	−	−	−	+	−	−	−	−	−	−	−
F_3	−	−	−	−	−	−	−	−	−	−	−	−
F_4	−	−	−	−	+	−	−	−	−	−	−	−
F_5	−	−	−	−	≈	−	−	−	−	−	−	−
F_6	−	−	−	−	+	−	−	−	−	−	−	−
F_7	−	−	−	−	+	−	−	−	−	−	−	−
F_8	−	−	−	−	−	−	−	−	−	−	−	−
F_9	−	−	−	−	−	−	≈	−	−	−	−	−
F_{10}	−	−	−	−	−	−	−	−	−	−	−	−
F_{11}	−	−	−	−	+	−	−	−	−	−	−	−
F_{12}	−	−	−	−	≈	−	−	≈	≈	≈	−	−
F_{13}	−	−	−	−	−	−	−	−	−	−	−	−
F_{14}	−	−	−	−	−	−	−	−	−	−	−	−
F_{15}	−	−	−	−	−	−	−	−	−	−	+	−

rank-sum test results are provided in Tables 4 and 5. In these figures and tables, the symbols "−", " + ", and "≈" indicate that the compared algorithm performs significantly worse, significantly better, or shows no significant difference from LeSTA at a significance level of 0.05, respectively.

From Fig. 3, it is evident that LeSTA attains the best average ranking across the 15 tested functions, underscoring its competitiveness against the other algorithms. Moreover, the Wilcoxon rank-sum test results in Fig. 4 and the detailed results in Tables 4 and 5 collectively reinforce that LeSTA significantly outperforms 11 algorithms in both 50D and 30D cases. The sole exception is the CEO algorithm, which exhibits notable competitiveness and achieves performance comparable to LeSTA on several functions. Nevertheless, LeSTA consistently delivers the best results and sustains the most stable performance across dimensions, emphasizing its robustness and superiority over all compared algorithms. These findings provide strong statistical evidence that LeSTA offers a competitive and reliable optimization capability.

4.4. Performance evaluation of dynamic economic dispatching

The dynamic economic dispatching (DED) problem in power systems is a nonlinear optimization problem with multiple constraints. Fuel cost is one of the objective functions in the DED problem. When considering the valve-point effects, the fuel cost of each thermal power unit can be modeled as a superposition of a quadratic function and a sinusoidal function. Therefore, the total fuel cost of all generating units over T dispatch periods can be expressed as:

$$\text{Minimize } F = \sum_{t=1}^{T}\sum_{i=1}^{N_G} F_{it}\left(P_{i,t}\right) \tag{9}$$

$$F_{it}\left(P_{i,t}\right) = a_i + b_i P_{i,t} + c_i P_{i,t}^2 + \left|d_i \sin\left(e_i\left(P_i^{\min} - P_{i,t}\right)\right)\right| \tag{10}$$

where a_i, b_i, c_i, d_i, and e_i denote the cost coefficients of the ith generator unit; $P_i^{\min}$ represents the minimum power output limit of the ith unit; and $P_{i,t}$ indicates the active power output of the ith unit during the dispatch period t. The equality and inequality constraints of the DED problem are presented as follows:

(1) Power balance constraints

$$\sum_{i=1}^{N_G} P_{i,t} - P_{load,t} - P_{loss,t} = 0, \quad t = 1, 2, \ldots, T \tag{11}$$

where $P_{load,t}$ and $P_{loss,t}$ represent the system load demand and network loss power at time period t, respectively. The calculation model for network loss power based on the B-coefficients method is given as follows:

$$P_{loss,t} = \sum_{t=1}^{N_G}\sum_{j=1}^{N_G} P_{i,t} B_{ij} P_{j,t} + \sum_{i=1}^{N_G} P_{i,t} B_{i0} + B_{00} \tag{12}$$

(2) Power operating constraints

$$P_i^{\min} \le P_{i,t} \le P_i^{\max} \tag{13}$$

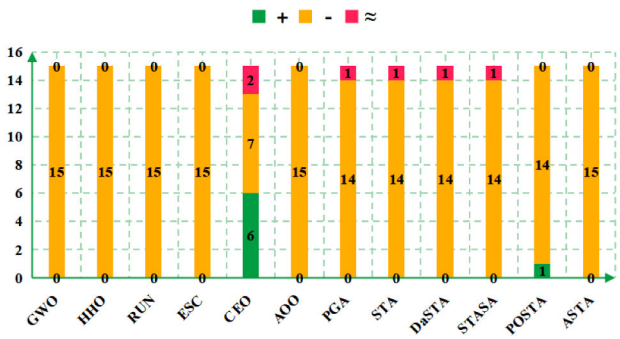
where $P_i^{\max}$ denotes the maximum power output of the ith generator unit.

Table 5
Wilcoxon rank-sum test results of LeSTA versus other algorithms on 50-dimensional functions.

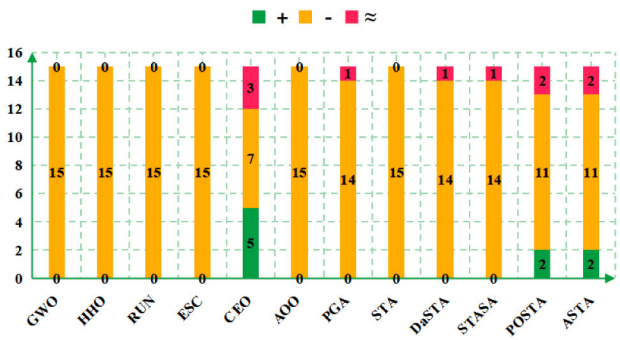
Func	GWO	HHO	RUN	ESC	CEO	AOO	PGA	STA	DaSTA	STASA	POSTA	ASTA
F_1	−	−	−	−	+	−	−	−	−	−	−	−
F_2	−	−	−	−	+	−	−	−	−	−	+	+
F_3	−	−	−	−	+	−	−	−	−	−	≈	≈
F_4	−	−	−	−	−	−	−	−	−	−	−	−
F_5	−	−	−	−	+	−	−	−	−	−	−	−
F_6	−	−	−	−	+	−	≈	−	−	−	−	−
F_7	−	−	−	−	≈	−	−	−	−	−	−	−
F_8	−	−	−	−	≈	−	−	−	−	−	+	+
F_9	−	−	−	−	−	−	−	−	−	−	−	−
F_{10}	−	−	−	−	−	−	−	−	−	−	−	−
F_{11}	−	−	−	−	−	−	−	−	−	−	−	−
F_{12}	−	−	−	−	−	−	−	−	−	≈	−	≈
F_{13}	−	−	−	−	−	−	−	−	≈	−	≈	−
F_{14}	−	−	−	−	≈	−	−	−	−	−	−	−
F_{15}	−	−	−	−	−	−	−	−	−	−	−	−

Table 6
Comparison of optimization algorithms.

Methods	Minimum cost	Average cost	Maximum cost	Standard deviation	Success rate (%)
GWO	1,047,524,195	6,403,210,878	11,625,679,109	3,632,102,125	0
HHO	15,616,011,722	23,479,107,305	34,743,470,868	6,915,077,596	0
RUN	66,172,283,180	88,265,416,584	115,182,509,138	13,909,550,081	0
ESC	124,303,468,811	161,736,140,858	191,275,473,478	19,881,844,003	0
CEO	2,913,625	2,933,652	2,961,449	17,044	100
AOO	3,002,896	1,418,405,364	4,165,348,941	1,476,522,472	20
PGA	2,965,185	3,586,107,774	6,580,217,666	2,246,946,802	10
STA	150,891,154	2,991,819,993	6,094,258,649	2,418,969,216	0
DaSTA	2,958,260	643,954,481	1,858,595,030	774,453,227	40
STASA	2,948,535	1,071,226,040	7,434,705,922	2,312,666,064	60
POSTA	2,939,116	1,003,113,689	4,299,237,525	1,645,833,398	40
ASTA	2,870,865	772,424,486	3,277,541,777	1,122,967,108	60
LeSTA	2,865,060	2,914,494	2,952,183	25,754	100



(a) 30D



(b) 50D

Fig. 4. Rank sum test results of twelve algorithms on fifteen test functions. (a)30D. (b)50D.

(3) Generating unit ramp rate constraints

$$\begin{cases} P_{i,t} - P_{i,t-1} \le U_{Ri} \times \Delta T \\ P_{i,t-1} - P_{i,t} \le D_{Ri} \times \Delta T \end{cases}, \quad i = 1, 2, \ldots, N_G \tag{14}$$

where ΔT denotes the duration of a dispatch period, while D_{Ri} and U_{Ri} represent the load reduction rate limit and load increase rate limit of the ith generator unit, respectively.

To validate the performance of the LeSTA algorithm in solving the DED problem, this experiment investigates a typical test system comprising 10 thermal power units. The experimental setup adopts a 24-h dispatch cycle with a 1-h time resolution. The physical parameters of each unit (including cost coefficients and ramping rates), time-series load forecast values, and network loss parameters are detailed in Basu (2008). Regarding the parameter configuration, the maximum number of function evaluations was set to 30e4, while the other parameters remained the same as in the benchmark tests. All constraints are handled using the exterior penalty function method, where constraint violations are incorporated into the objective function as penalty terms. To evaluate algorithmic performance, other twelve algorithms are selected for comparison, with each algorithm executing 30 independent trials to mitigate random fluctuations. Table 6 presents the statistical comparison results of the thirteen algorithms, including minimum, mean, maximum, standard deviation values, and success rate. Here, the success rate is defined as the percentage of runs in which an algorithm successfully finds a feasible solution that satisfies all constraints, calculated as the ratio of successful runs to the total number of 30 independent trials.

Based on the comprehensive statistical results presented in Table 6, the proposed LeSTA algorithm demonstrates superior performance in solving the DED problem, outperforming all compared algorithms in multiple key metrics. LeSTA achieves the minimum cost value of 2,865,060 among all compared algorithms, indicating its

exceptional capability to locate high-quality solutions in the complex non-convex search space characterized by valve-point effects. The algorithm maintains remarkable solution consistency with an average cost of 2,914,494 and substantially low standard deviation of 25,754. While the CEO algorithm also achieves a 100 % success rate, both its minimum cost (2,913,625) and average cost (2,933,652) are higher than those of LeSTA. Similarly, although ASTA shows competitive minimum cost (2,870,865), its significantly higher average cost (772,424,486) and standard deviation (1,122,967,108) reveal poor stability compared to LeSTA's consistent performance.

The 100 % success rate achieved by LeSTA is particularly noteworthy when contrasted with the complete failure (0 % success rate) of several established algorithms including GWO, HHO, RUN, ESC, and STA. This perfect success rate, combined with superior cost values, highlights LeSTA's robustness in handling complex DED constraints through the exterior penalty function method. The exceptional performance can be attributed to LeSTA's effective balance between exploration and exploitation capabilities, enabling efficient navigation through the challenging non-convex, non-smooth search space while systematically handling various system constraints. The results indicate that LeSTA performs effectively for the DED problem, with clear advantages observed in solution quality, reliability, and success rate. This performance highlights its potential for application in practical dynamic economic dispatch within modern power systems.

5. Conclusion

This paper proposed the LeSTA, a population-based extension of the standard STA. Unlike the original STA that primarily relies on self-learning, LeSTA integrates three additional learning strategies, namely peer-learning, best-learning, and experience-learning, together with self-learning in a unified framework. In addition, corresponding state transformation operators and a crossover mechanism were designed to explicitly balance exploration and exploitation, strengthen inter-individual knowledge transfer, and alleviate premature convergence.

Comprehensive experiments on 15 benchmark functions demonstrated that LeSTA consistently achieves superior performance compared to other 12 metaheuristic algorithms. It exhibits a strong capability in escaping deceptive local optima, a critical ability for navigating the complex search space of the non-convex DED problem, while maintaining general robustness essential for practical use.

Nevertheless, it is important to note a limitation of the current LeSTA: the four learning mechanisms are applied in a fixed, sequential manner, which may not achieve optimal search efficiency. The potential for dynamically selecting these mechanisms based on the search state remains an open question for future investigation.

Future research will therefore focus on the development of adaptive operator selection mechanisms, alongside theoretical convergence analysis, extensions to discrete and combinatorial optimization problems, and applications to computationally expensive real-world scenarios, such as large-scale sensor deployment and hyperparameter optimization for deep neural networks. These directions are expected to further enhance both the applicability and the theoretical foundation of LeSTA.

CRediT authorship contribution statement

Yingchao Dong: Writing – review & editing, Conceptualization, Investigation, Software, Visualization; **Ge Lan:** Writing – review & editing, Validation, Supervision; **Xiaojun Zhou:** Writing – review & editing, Conceptualization, Supervision, Software.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant No. 62573430, 62273357), the Natural Science Foundation of Xinjiang Uygur Autonomous Region (Grant No. 2025D01B92).

References

Ahmadianfar, I., Heidari, A. A., Gandomi, A. H., Chu, X., & Chen, H. (2021). Run beyond the metaphor: An efficient optimization algorithm based on runge kutta method. *Expert Systems with Applications*, *181*, 115079.

Basu, M. (2008). Dynamic economic emission dispatch using nondominated sorting genetic algorithm-II. *International Journal of Electrical Power & Energy Systems*, *30*(2), 140–149.

Bohat, V. K., Hashim, F. A., Batra, H., & Elaziz, M. A. (2025). Phototropic growth algorithm: A novel metaheuristic inspired from phototropic growth of plants. *Knowledge-Based Systems*, *322*, 113548.

Derrac, J., García, S., Molina, D., & Herrera, F. (2011). A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms. *Swarm and Evolutionary Computation*, *1*(1), 3–18.

Dong, Y., Wang, C., Zhang, H., & Zhou, X. (2024). A novel multi-objective optimization framework for optimal integrated energy system planning with demand response under multiple uncertainties. *Information Sciences*, *663*, 120252.

Dong, Y., Zhang, H., Ma, P., Wang, C., & Zhou, X. (2023a). A hybrid robust-interval optimization approach for integrated energy systems planning under uncertainties. *Energy*, *274*, 127267.

Dong, Y., Zhang, H., Wang, C., & Zhou, X. (2021a). A novel hybrid model based on bernstein polynomial with mixture of gaussians for wind power forecasting. *Applied Energy*, *286*, 116545.

Dong, Y., Zhang, H., Wang, C., & Zhou, X. (2021b). Wind power forecasting based on stacking ensemble model, decomposition and intelligent optimization algorithm. *Neurocomputing*, *462*, 169–184.

Dong, Y., Zhang, H., Wang, C., & Zhou, X. (2022). An adaptive state transition algorithm with local enhancement for global optimization. *Applied Soft Computing*, *121*, 108733.

Dong, Y., Zhang, H., Wang, C., & Zhou, X. (2023b). Robust optimal scheduling for integrated energy systems based on multi-objective confidence gap decision theory. *Expert Systems with Applications*, *228*, 120304.

Dong, Y., Zhang, S., Zhang, H., Zhou, X., & Jiang, J. (2025). Chaotic evolution optimization: A novel metaheuristic algorithm inspired by chaotic dynamics. *Solitons & Fractals*, *192*, 116049.

Dorigo, M., Birattari, M., & Stutzle, T. (2007). Ant colony optimization. *IEEE Computational Intelligence Magazine*, *1*(4), 28–39.

Du, Y., Zhou, X., Yang, C., & Huang, T. (2023). An interactive feature selection method based on multi-step state transition algorithm for high-dimensional data. *Knowledge-Based Systems*, *282*, 111102.

Han, X., Dong, Y., Yue, L., & Xu, Q. (2019). State transition simulated annealing algorithm for discrete-continuous optimization problems. *IEEE Access*, *7*, 44391–44403.

Heidari, A. A., Mirjalili, S., Faris, H., Aljarah, I., Mafarja, M., & Chen, H. (2019). Harris hawks optimization: Algorithm and applications *Future Generation Computer Systems* (*vol. 97*), 849–872

Holland, J. H. (1992). Genetic algorithms. *Scientific American*, *267*(1), 66–73.

Kudela, J. (2022). A critical problem in benchmarking and analysis of evolutionary computation methods. *Nature Machine Intelligence*, *4*(12), 1238–1245.

Lin, F., Zhou, X., Li, C., Huang, T., & Yang, C. (2022). Data-driven state transition algorithm for fuzzy chance-constrained dynamic optimization. *IEEE Transactions on Neural Networks and Learning Systems*, *34*(9), 5322–5331.

Mirjalili, S., Mirjalili, S. M., & Lewis, A. (2014). Grey wolf optimizer. *Advances in engineering software*, *69*, 46–61.

Ouyang, K., Fu, S., Chen, Y., Cai, Q., Heidari, A. A., & Chen, H. (2024). Escape: An optimization method based on crowd evacuation behaviors. *Artificial Intelligence Review*, *58*(1), 19.

Pereira, D. G., Afonso, A., & Medeiros, F. M. (2015). Overview of friedman's test and post-hoc analysis. *Communications in Statistics-Simulation and Computation*, *44*(10), 2636–2653.

Storn, R., & Price, K. (1997). Differential evolution-a simple and efficient heuristic for global optimization over continuous spaces. *Journal of Global Optimization*, *11*, 341–359.

Venter, G., & Sobieszczanski-Sobieski, J. (2003). Particle swarm optimization. *AIAA Journal*, *41*(8), 1583–1589.

Wang, R.-B., Hu, R.-B., Geng, F.-D., Xu, L., Chu, S.-C., Pan, J.-S., Meng, Z.-Y., & Mirjalili, S. (2025). The animated oat optimization algorithm: A nature-inspired metaheuristic for engineering optimization and a case study on wireless sensor networks. Knowledge-Based Systems.

Zhou, X., Li, M., Du, Y., Yang, C., & Wen, S. (2025). Interpretable multiobjective feature selection via visualization in froth flotation process. *IEEE Transactions on Industrial Informatics*, *21*(3), 2530–2539.

Zhou, X., Shi, P., Lim, C.-C., Yang, C., & Gui, W. (2018). A dynamic state transition algorithm with application to sensor network localization. *Neurocomputing*, *273*, 237–250.

Zhou, X., Sun, Y., Huang, Z., Yang, C., & Yen, G. G. (2022). Dynamic multi-objective optimization and fuzzy ahp for copper removal process of zinc hydrometallurgy. *Applied Soft Computing*, *129*, 109613.

Zhou, X., Tang, Z., Wang, N., Yang, C., & Huang, T. (2024). A novel state transition algorithm with adaptive fuzzy penalty for multi-constraint UAV path planning. *Expert Systems with Applications*, *248*, 123481.

Zhou, X., Yang, C., & Gui, W. (2012). State transition algorithm. *Journal of Industrial and Management Optimization*, *8*(4), 1039–1056.

Zhou, X., Yang, C., & Gui, W. (2019). A statistical study on parameter selection of operators in continuous state transition algorithm. *IEEE transactions on cybernetics*, *49*(10), 3722–3730.